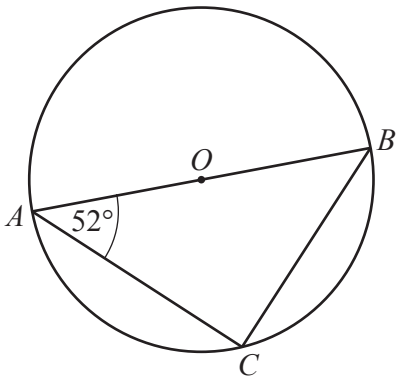
0580_s21_qp_33_Q6

6 **(a)**



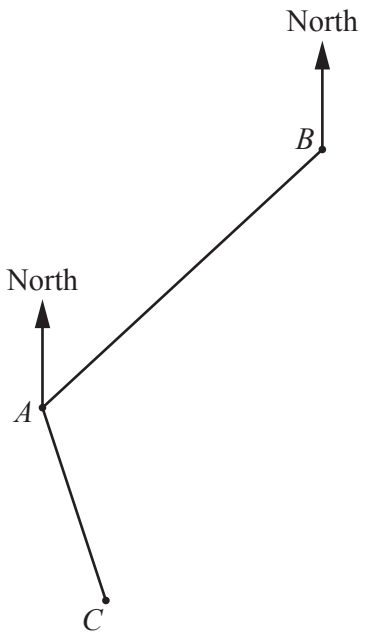
NOT TO SCALE

AB is the diameter of a circle, centre *O*.
C is a point on the circle and angle *BAC* = 52°.

Find angle *ABC*.

Angle *ABC* = ... [2]

(b) The diagram shows the positions of town *A*, town *B* and town *C*.



NOT TO SCALE

The bearing of town *B* from town *A* is 042°.
The bearing of town *C* from town *A* is 146°.

(i) Find angle *BAC*.

Angle *BAC* = ... [2]

(ii) Find the bearing of town *A* from town *B*.

..... [2]

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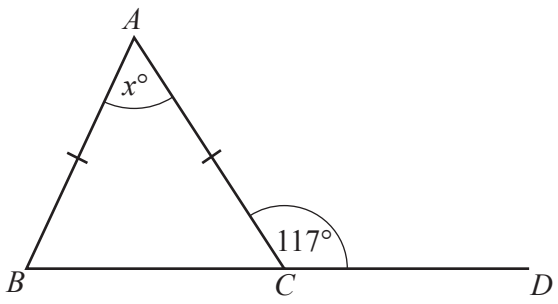
(c)



NOT TO SCALE

Triangle ABC is isosceles with $AB = AC$.
BCD is a straight line and angle $ACD = 117°$.

Find the value of x.

$x =$ ..... [3]